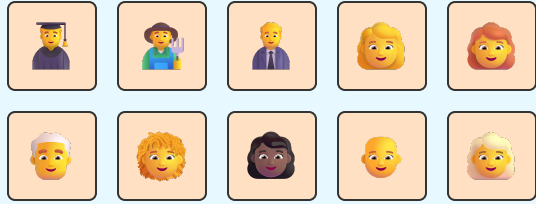


DynaMem-Bench Construction — Two-Phase Pipeline

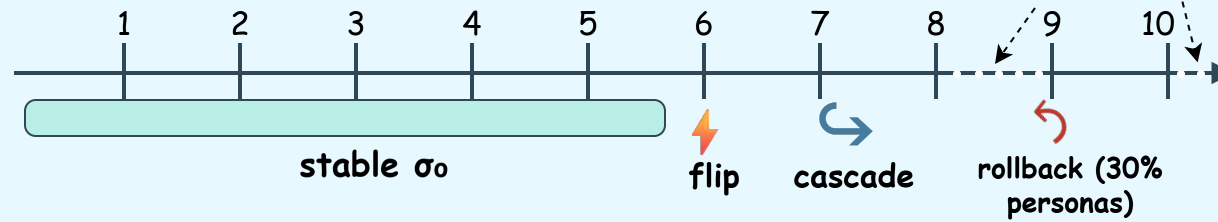
Phase 1: OFFLINE — system-agnostic, shared across all 7 tested systems

Personas



10 personas, each carrying a state schema with a main-core variable + auxiliary-core with cascade lag

State Trajectory $\sigma_0 \rightarrow \sigma_{10}$



★ **MUST-EXPOSE SCHEDULE** ★
same fact set pinned across all 7 systems for fairness

Questions + Oracle

? POS-fresh ? POS-tracking

? NEG (refuse-correct)

 C_{oracle} per Q

10 questions per (persona, session);
minimal oracle context supports the gold answer

Offline bundle: personas, state, must-expose, questions, oracle — identical input for every tested system

Phase 2: ON-POLICY — per-system session loop

For each (system, persona): reset memory M ; run 10 sessions

SUB-STEP 1

 turn 0: seed utterance (pinned from offline)



GPT-4o
user simulator

SUB-STEP 2
turns 1...N

conditioned on persona, current state σ_t , residual
must-expose list, dialogue history + system's last reply

* force-expose if must-expose residual is not exhausted by turn 9

tested memory
system

SUB-STEP 3



ReadOnlyAdapter

session-end probe — writes BLOCKED,
 M never polluted by evaluation traffic

962 evaluation asks (790 POS / 172 NEG)

same question set across all 7 systems — each system saw its own on-policy dialogue